

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

10 Semantics of Programming Languages (nk480)

Regular expressions are defined by the following grammar:

$r ::= c$	Matches the single-character word c
$ \epsilon$	Matches the empty word
$ r_1 \circ r_2$	Matches the concatenation of an r_1 -word and an r_2 -word
$ 0$	Matches no words
$ r_1 + r_2$	Matches any r_1 -word or r_2 -word
$ r^*$	Matches the concatenation of a finite number of r -words

Operational
Semantics

- (a) Give a set of inference rules defining a relation for when a word w is matched by a regular expression r . Use the notation $w \cdot w'$ to denote concatenation.

[8 marks]

Answer:

$$\begin{array}{c}
 \overline{c \text{ in } c} \qquad \overline{\epsilon \text{ in } \epsilon} \\
 \frac{w_1 \text{ in } r_1 \quad w_2 \text{ in } r_2}{w_1 \cdot w_2 \text{ in } r_1 \circ r_2} \quad \frac{w \text{ in } r_i \quad i \in \{1, 2\}}{w \text{ in } r_1 + r_2} \\
 \overline{\epsilon \text{ in } r^*} \quad \frac{w_1 \text{ in } r \quad w_2 \text{ in } r^*}{w_1 \cdot w_2 \text{ in } r^*}
 \end{array}$$

Note that there is no rule for 0.

Semantic
Equivalence

- (b) (i) Using the matching relation defined above, define a suitable notion of semantic equivalence $r_1 \simeq r_2$ for regular expressions. [4 marks]
- (ii) Use this definition to prove that $(r + r') \simeq (r' + r)$. You may use inversion lemmas without proof, as long as they are explicitly indicated. [4 marks]

Answer: We can define $r_1 \simeq r_2$ as “for all words w , $w \text{ in } r_1$ if and only if $w \text{ in } r_2$ ”.

We can use this to prove that $r + r' \simeq r' + r$ as follows.

- (i) Assume a word w .
- (ii) Without loss of generality, it suffices to prove the left-to-right direction.
- (iii) Assume that $w \text{ in } r + r'$.
- (iv) By inversion on the matching relation, we know that either $w \text{ in } r$, or $w \text{ in } r'$.
- (v) In the case that $w \text{ in } r$, we can use an inference rule to conclude that $w \text{ in } r' + r$.
- (vi) Symmetrically, if $w \text{ in } r'$, we can use an inference rule to conclude that $w \text{ in } r' + r$.

In order to get full marks for this question, students have to supply a reasonable definition of semantic equivalence, and also supply a correct proof. If the definition in (b)(i) is wrong, then it is still possible to get partial marks for (b)(ii) if a correct, nontrivial proof is given.

Inductive
Definitions

- (c) Define an inductive relation $r \text{ null}$ characterizing the regular expressions r for which $\epsilon \text{ in } r$. [4 marks]

Answer:

$$\frac{\overline{\epsilon \text{ null}}}{\frac{r_i \text{ null}}{r_1 + r_2 \text{ null}} \quad i \in \{1, 2\}} \quad \frac{\frac{r_1 \text{ null}}{r_1 \circ r_2 \text{ null}} \quad \frac{r_2 \text{ null}}{\text{null}}}{\overline{r * \text{ null}}}$$
